SAKTHIMURUGAN V

Electronics and Communiation Engineer

+91 9789378758 | Tiruvannamalai, Tamil Nadu | sakthimuruganvenkat@gmail.com

LinkedIn | GitHub | Portfolio

CAREER SUMMARY

ECE graduate with hands-on experience in VLSI, RTL Design, Verilog HDL, FPGA development, and digital design. Skilled in RTL coding, simulation, synthesis, and hardware implementation using Artix-7, Spartan-6, and BASYS 3 FPGAs. Strong understanding of digital logic, FSMs, timing concepts, and ASIC design flow, with knowledge of UART, SPI, I²C, and CAN. Seeking roles in VLSI, RTL Design, FPGA, or ASIC Design.

SKILL STACK

Programming Language	:	Verilog HDL.
Digital Design	:	Digital Electronics, RTL Design, FSM Design, Simulation and Synthesis.
AMBA Protocols	:	AXI, AHB, APB.
FPGA Platforms	:	Spartan-3E, Spartan-6, Artix-7, BASYS 3.
ASIC Design & Verification	:	ASIC Design Flow, UVM Concepts, Testbench Development, Simulation and Debugging.
Tools & IDEs	:	AMD Vivado, Xilinx ISE, ModelSim, Icarus Verilog, GTKWave.
Soft Skills	:	Problem-solving, Teamwork, Adaptability, Communication, Time Management.

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering 2022 – 2026 | CGPA: 8.5
Adhiparasakthi Engineering College, Melmaruvathur.

Higher Secondary (XII) – SRI International Matric Hr. Sec. School, Birudur – (2022) | Percentage: 85.5%

Secondary (X) – SRI International Matric Hr. Sec. School, Birudur – (2020) | Percentage: 91.4%

PROJECTS

Traffic Light Controller (FSM Design)

- Designed and developed a parameterized 4-bit Arithmetic Logic Unit (ALU) using Verilog HDL, supporting scalable RTL architecture for arithmetic and logical operations.
- Developed comprehensive Verilog testbenches and performed functional simulation, waveform analysis, and RTL verification to ensure design correctness.
- Executed RTL synthesis, implementation, and hardware validation on an AMD Artix-7 FPGA using Xilinx Vivado, verifying the synthesized design on physical hardware.

FSM-Based Traffic Light Controller Using Verilog HDL

- Designed and developed an FSM-based Traffic Light Controller using Verilog HDL, implementing automated traffic signal sequencing with defined state transitions.
- Developed a synchronous sequential RTL architecture with clock-driven state management and control logic for reliable signal operation.
- Performed RTL simulation and synthesis, followed by hardware implementation and validation on a Xilinx Spartan-6 FPGA using Xilinx ISE.

INTERNSHIP EXPERIENCE

Hardware Testing Intern – Vi Microsystems Pvt. Ltd., Chennai — Jan 2026 - Apr 2026

- Gained hands-on experience in FPGA-based RTL design and digital system development using Basys 3, Xilinx Artix-7, and Spartan-6 development boards.
- Designed and implemented RTL modules using Verilog HDL, followed by functional simulation, synthesis, implementation, timing analysis, and bitstream generation for FPGA deployment.
- Programmed and validated embedded platforms including 8051, 8085, 8086, Arduino, ESP32, Raspberry Pi, and Raspberry Pi Pico, while performing hardware debugging and peripheral-level testing.

Embedded Systems Intern – National Small Industries Corporation, Chennai — July 2025 - Aug 2025

- Implemented sensor interfacing using ADC and GPIO for data acquisition and device control.
- Worked with UART, SPI, and I²C protocols for communication between controllers, sensors, and peripherals.
- Performed basic debugging and troubleshooting of embedded hardware, firmware, and communication issues.

CERTIFICATIONS

- System Design through Verilog, NPTEL
- Embedded Systems Design, NPTEL
- Introduction to Embedded Systems and Internet of Things, Coursera
- Embedded Systems Using C, EDUCBA